Advance Python Assignment

Topic-7: Inheritance

Q1. Explain different types of inheritance in Python.

Ans.
Inheritance in Python allows one class to use the properties and methods of another class. It helps in reducing code duplication and makes programs more efficient. There are different types of inheritance such as single, multilevel, multiple, hierarchical, and hybrid inheritance. In single inheritance, one child class inherits from one parent class. In multilevel inheritance, a class inherits from another derived class. Multiple inheritance allows one class to inherit from more than one class. Hierarchical inheritance means many child classes inherit from one parent class. Hybrid inheritance is a combination of different inheritance types used together.

Example:
```
class Animal:
    def sound(self):
        print("Animal makes sound")

class Dog(Animal):
    def bark(self):
        print("Dog barks")

d = Dog()
d.sound()
d.bark()
```

Q2. Explain the super() function in Python inheritance.

Ans.
The super() function in Python is used to call methods and constructors of the parent class from the child class. It helps in avoiding direct use of the parent class name and makes the code more flexible. Using super(), we can extend the functionality of the parent class without rewriting the same code. It is especially useful when working with inheritance and constructors in object-oriented

programming.

Example:

```
class Base:
    def __init__(self):
        print("Base class constructor")

class Derived(Base):
    def __init__(self):
        super().__init__()
        print("Derived class constructor")

obj = Derived()
```